Issues or Traits? Understanding American Partisan Favorability and Animosity through Open-Ended Survey Responses

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1 Executive Summary

Democratic and Republican party supporters increasingly dislike each other, but reasons for this rising animosity has been under debate.

Over two studies, we investigate how the content of what Democrats and Republicans like and dislike about each party predicts their favorability toward their own party and their animosity toward the other party. We further analyze the generalizability of our findings. We use open-ended survey responses from five cross-sectional samples of the American National Election Studies (2008-24) to gauge party favorability content. We combine a skip-gram text mining approach with relaxed LASSO to analyze open-ended responses' predictiveness of party favorability. To analyze generalizability, we conduct relaxed LASSO using a range of demographic variables and political attitudes to predict non-response to the ANES open-ended survey questions.

We find that Americans primarily use issue-based words to explain why they like and dislike each party. Democrats and Republicans appear not to dislike each other personally simply because they support the opposite party. On the opposite, such dislike seems to stem from substantive policy disagreements.

Caution is needed when interpreting our findings, as we find substantial selection bias in the open-ended survey responses. Our findings skew toward people who are politically interested, highly educated, male, white, and partisan.

2 Introduction and Motivation

One need not be a keen observer of American politics to know that supporters of the Democratic and Republican parties have increasingly disliked each other over the past 20 years. Increasingly, Americans seem to be unable to find common ground across partisan lines. This phenomenon of political polarization has led many to worry about a decline in American democracy, where people become less willing to find political compromise with members of the opposite party, less supportive of democratic norms, such as the independence of each of the three branches of government, and more likely to engage in political violence against out-party supporters.

Nevertheless, two competing accounts about why Americans dislike out-party supporters offer competing implications about how intractable political polarization is in the American public. One camp of political science scholars argue that Americans dislike members of the out-party simply because of their political identity. Scholars who term this affective polarization (e.g., Iyengar, Sood, and Lelkes, 2012) argue that increasingly, out-party animosity functions similarly as racism and religious discrimination, where people in competing social groups dislike each other automatically, even when they do not disagree on policy opinions. There appears to be plenty of empirical support for affective polarization. According to Iyengar and colleagues' analysis of survey data since the 1960s, Americans report being increasingly displeased if their children were to marry someone from another party. Over the same timespan, they have also become increasingly likely to rate supporters of their own party as intelligent, and supporters of the opposite party as selfish. This constitutes what we here call the trait-based account of American political polarization, where politics becomes personal, and animosity is automatic.

In contrast, the issue-based account takes the same evidence of increasing partisan dislike, and offers a competing argument. Americans care more about policy opinion disagreements than differences in partisan identity, this account argues, and mostly use partisanship to infer other people's substantive political issues positions (Orr and Huber, 2020). It just so happens that over the recent decades, the Democratic and Republican Party have moved further apart on their issue positions, such that you would be hard-pressed to find any pro-welfare Republican politicians or any anti-abortion Democratic politicians (Levendusky, 2009). In other words, a Democrat may still feel deeply troubled by their daughter marrying a Republican supporter, but not simply because he is a Republican per se, but because it is highly likely that the husband is anti- abortion. Under this account, political polarization becomes less normatively problematic. Party supporters debate one another vigorously based on substantive policies rather than social identities, which is a natural part of how democracies resolve disagreements.

In the current project, we attempt to offer some descriptive evidence to arbitrate between these two competing accounts of American political polarization. We achieve this by analyzing why Americans say they like and dislike the Democratic and Republican party. Through examining the content of Americans' self-reported out-party animosity and in-party favorability, we hope to understand whether issues or traits predominantly predict whether Americans like or dislike each party.

3 Data Description

We utilize closed-ended and open-ended survey responses from the American National Election Studies (ANES), one of the definitive data sources for American politics analysts. The data set has several benefits for our analysis. One, it has a large sample size ranging from 2,300 to 8,400 in our years of interest. Personally collecting such a large sample would have been unrealistically costly for us. Thankfully, ANES has made most of its data freely available for research purposes. Two, ANES applies probabilistic sampling for each year that it fields its survey. Since we hope to make population-level inferences about Americans in general, a probabilistic sample gives us the inferential power as each American resident theoretically has equal probability of being contacted by the survey administrators. Third, and most importantly for our purposes, ANES asks many questions about political attitudes and demographics in the exact same way over many consecutive waves. This consistency in question wording allows us to pool and compare data from different years. In contrast, any survey question that measures the same concept but words the question differently, raises the concern that such wording changes may systematically alter responses.

Specifically, we used the fresh cross-sectional pre-election samples from 2008 to 2024 for our analysis, resulting in a pooled sample size of N=26,307. These combine surveys conducted in both face-to-face and online formats. ANES surveys are generally fielded on election years. Thus, this gave us five years of data (2008, 12, 16, 20, 24) that traced as far back as the start of the first Obama administration, when political polarization was high but in no way comparable to today's levels. The year 2008 was set as our starting year because this was the first year in which ANES made its open-ended survey response data publicly available. Our main outcome of interest is the feeling thermometer scale for the Democratic and the Republican party, a common measure in American politics literature for partisan favorability and animosity. Respondents were instructed to rate how warmly they feel about each party on a 0-100 scale. To quote from the ANES codebook, "Ratings between 50 degrees-100 degrees mean that you feel favorably and warm toward the group; ratings between 0 and 50 degrees mean that you don't feel favorably towards the group and that you don't care too much for that group. If you don't feel particularly warm or cold toward a group you would rate them at 50 degrees."

Our main predictors are extracted from open-ended responses to four questions: "What is it that you like [dislike] about the Democratic [Republican] party?" This question follows an earlier binary response question asking, "Is there anything that you like [dislike] about the Democratic [Republican] party?" Respondents who replied "No" would not have reached the open-ended question. As selection bias in open-ended survey responses is a known issue in public opinion research, we included this binary variable to analyze predictors of non-response to our open-ended questions.

To further probe selection bias, we included a range of multiple-choice demographic variables and political attitude questions also available in the ANES data we selected. Our variable selection criteria were that 1) each variable must have been measured in all five waves of our data set, and 2) each variable must plausibly be substantive relevant for our research question. A description of each of these closed-ended, multiple-choice variables can be found in `anes_timeseries_cdf_codebook_Varlist.pdf`, where we highlighted all variables we selected.

Two other data pre-processing steps are worth noting here, should the reader wish to analyze ANES data in the future. First, data merging requires extra caution. ANES stored its open-ended data files in separate spreadsheets from each year's main data file, as well as from what it calls the cumulative data file, which compiles all the closed-ended, multiple-choice responses. Merging the closed- and open-ended data sets require matching on the respondent ID numbers. However, in some cases, errors in survey administration makes this process difficult. In particular, in 2016, every respondent in the main data file was given a second set of response IDs for unclear reasons, whereas the open-ended data file included only the original, first set of response IDs. This thus requires creating a mapping and matching function of the two sets of response IDs in preparation for merging. Second, NA's were coded in diverse ways – as 98 or 99 for some variables, -7 or -9 for others, 'Inapplicable' for some others, and simply NA for still others. Because certain variables take values such as 98 or 99 as valid responses (ex. the 0-100 continuous feeling thermometer measures), extra care is required when coding for NA. The only way to properly recode the NAs was by referencing the codebook for each included variable's coding schemes. Because we included a total of 37 closed-ended variables, here we used Claude Code to help us scrape the codebook and write the recoding script to accelerate the process.

4 Exploratory Data Analysis

In this section, we describe three features of our data: 1) sample composition by year, 2) how feeling thermometer values, our main outcome, has changed from 2008 to 2024, and 3) non-response distributions.

Sample composition by year One concern with analyzing any pooled data collected from different times is that time-variant variables may affect statistical inference. This concern is justified with our present data set. As the left-hand-side figure below shows, sample size varies widely by year. Pooling the data without accounting for time-variant sample size changes may effectively overweight years that happened to have a larger sample size (ex. 2020). Additionally, as the right-hand-side figure below shows, the partisan composition of each year's sample has changed considerably over time, where there was a declining proportion of Democrats and independents, and a rising proportion of Republicans and lean Republicans. It is hard to disentangle whether such year-to-year differences are due to changes in survey administration methods, changes in the American electorate in the real world, or some combination of both. Nevertheless, the presence of these idiosyncrasies from year to year motivates us to use year fixed effects for all our statistical models in subsequent analyses.

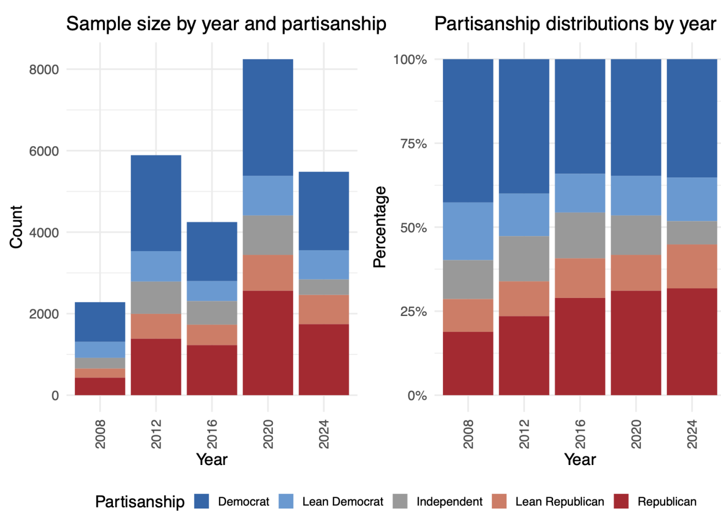

Figure 1: Longitudinal Trends of Sample Size and Partisanship Distributions.

Feeling thermometer ratings, 08-24 Out-party feeling thermometer results largely correspond to what is identified in prior literature. From 2008 to 2024, Americans increasingly dislike the out-party (left-hand- side), while in-party feelings have remained steady.

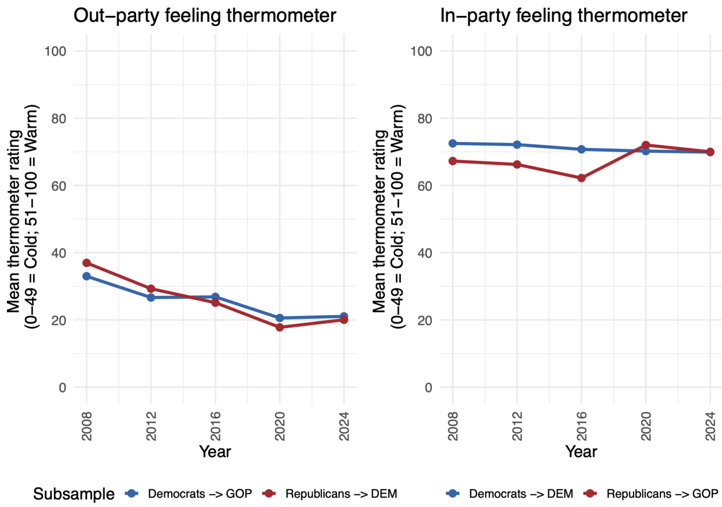

Figure 2: In- and Out-Party Feeling Thermometer

Examining the full sample, feelings for the Democratic party has declined over time, driven mostly by Republican and independents' animosity. Additionally, feelings for the Republican party has remained steady, such that Democrats' and independents' decreasing warmth is canceled out by Republicans' increasing warmth toward their own party.

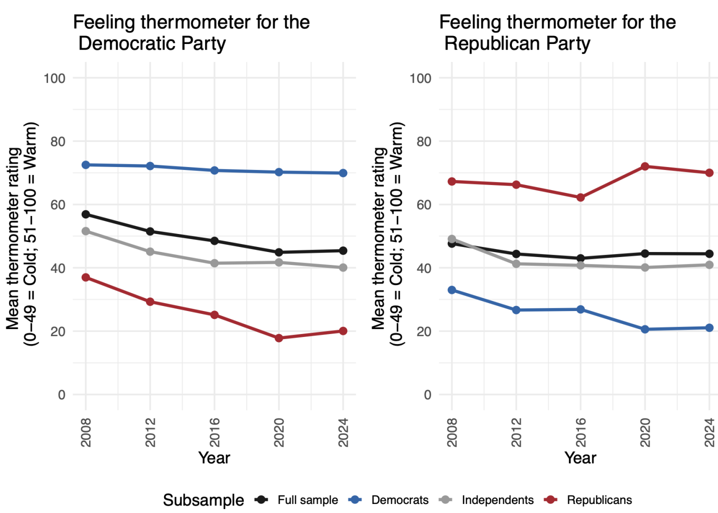

Figure 3: Feeling ThermometerS For the Two Parties

Distribution of nonresponses Depending on the question asked, 40-60% of the sample was not included in our analysis. Most respondents did not respond because they did not reach the open-ended question, because they indicated they did not have anything they liked or disliked about either party. However, we also identified two other sources of non-response. Some did not respond to the parent binary question in the first place. Others reached the open-ended question but did not give a valid response. The significant non-response rate motivated us to analyze our selection bias. In Study 2 of our main analysis, as will be elaborated on subsequently, we analyze what demographics and political attitudes predict non- response to the open-ended questions.

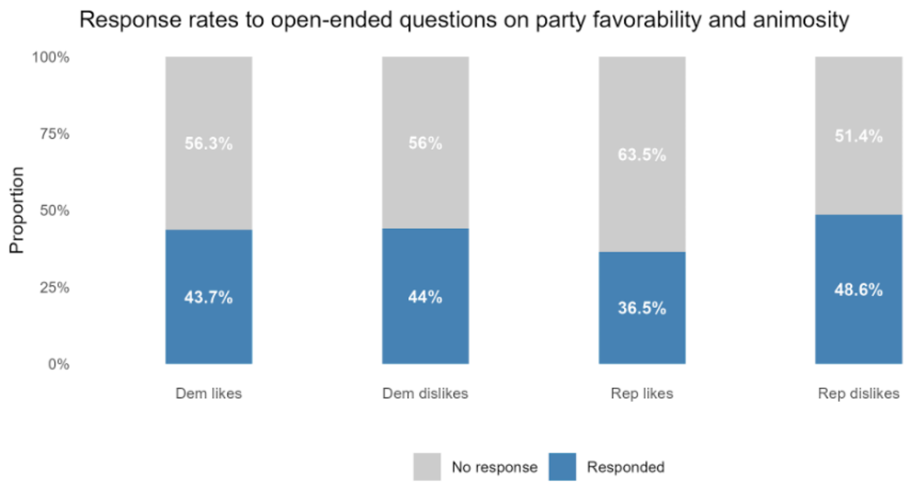

Figure 4: Response Rates to Open-Ended Questions on Party Favorability and Animosity

5 Methods

5.1 Study 1: Text Mining

Outcome Variables: Feeling Thermometers Our main outcome variables (DVs) for Study 1 are feeling thermometers which operationalize Democrat and Republican favorability and animosity. The American National Election Studies (ANES) asks respondents to rate each political party on a feeling thermometer scale from 0 to 100, where 0 represents the most negative feeling and 100 the most positive. Separate thermometers are provided for the Democratic and Republican parties. Figure 1 presents the empirical cumulative distribution functions (CDFs) for both thermometers across the pooled sample.

Both distributions (see Appendix S1G1) cluster at stepped intervals around round numbers, which is a well-documented feature of ANES feeling thermometers. The median thermometer rating across both parties, marked by the red line, is 50. Most notably, 15–20% of respondents assign a score of exactly 0 to each party, indicating strong animosity. This clustering, combined with the fact that the scale is bounded and not normally distributed, makes a continuous outcome inappropriate for regression modeling.

We therefore recode both thermometers into five ordered classes capturing meaningful differences in partisan affect. A five-class outcome is preferred over a binary split because it preserves substantive distinctions between peak (0–19) and moderate animosity (20–39), and between peak (80–100) and moderate favorability (60–79).

All five classes are well-populated, ranging from approximately 8% (unfavorable to Democrats) to 27% (highly unfavorable for Republicans), confirming that the recoding retains sufficient observations for multinomial modeling (see Appendix S1T1). We also tested an alternative outcome measure: the CSES party liking scale (0–10). This found substantively consistent results; see

Appendix A1.

Predictors: Open-Ended Survey Responses ANES respondents are asked four open-ended questions, providing text responses on why they like or dislike each party. For each respondent, we combine the Democratic like and dislike responses into a single `dem_text` column, and the Republican like and dislike responses into `rep_text`. Pooling likes and dislikes within each party provides a single, richer corpus of opinions about each party per respondent, and avoids modeling the same underlying affective orientation twice in separate models. The Democrat corpus is then used to predict Democrat thermometer ratings, and the Republican corpus to predict Republican thermometer ratings.

Approximately 40-60% of respondents provided no usable open-ended text for either party. These non-respondents are excluded from the text mining models but retained in the full dataset for Study 2 (RQ2), which specifically examines the predictors of non-response. Binary flags (`has_dem_text`, `has_rep_text`) are created for this purpose. The models are estimated on the full respondent sample, i.e., including both in-party and out-party raters, rather than restricting to out-party raters only. This choice preserves the full range of thermometer variation and avoids discarding independent and cross-partisan respondents whose evaluations may be informative. A partisan-stratified analysis restricting to out-party raters is reported as a robustness check in Appendix A2.

Text Representation: Skip-Grams We represent each open-ended response as a document-term matrix (DTM) using 2-skip-2-grams: word pairs within a window of up to two intervening words. For example, "they are not corrupt" generates `not_corrupt` as a feature despite the intervening word "are" — correctly capturing the negation that a bigram would miss. This is preferred over unigrams or bigrams because it handles negation and qualified language common in conversational political text (e.g., "not really socialist", "never truly fair") without requiring full sentence parsing.

Text pre-processing removes stopwords, numbers, and converts to lowercase. Punctuation is retained to preserve sentence-level adjacency information. All DTMs are filtered to retain only terms appearing in at least 1% of documents, yielding 220 terms for the Democrat thermometer model, and 207 terms for the Republican model, corresponding to sparsity of 97–98% across models. This is consistent with what is expected for short, open-ended survey responses such as these. Stemming and n-gram representations were also tested and produced substantively consistent results, as presented in Appendix A3.

Multinomial LASSO We use multinomial LASSO to select which terms most strongly predict our DVs. We use `cv.glmnet` with `alpha = 0.99`, an elastic net close to pure LASSO, with a tiny 'Ridge' component to assist with multicollinearity. LASSO is appropriate here because the number of candidate terms (200+) greatly exceeds what a standard logistic regression could handle without regularization. Our primary goal is intepretation of the selected terms.

We force in survey year (`year`) as a factor variable with zero penalty to its coefficients. This ensures that year fixed effects are included in every model, controlling for a documented increase in partisan polarization over 2008–2024. Consequently, coefficients on text terms reflect the relationship between language and thermometer ratings within years rather than being confounded by between year effects. Year fixed effects span 2008, 2012, 2016, 2020, and 2024, with 2008 as the reference category.

We use `lambda.1se` as our regularization parameter, i.e., the largest lambda within one standard error of the minimum cross-validated misclassification error. This produces a more parsimonious model than `lambda.min` while maintaining comparable predictive performance. Wald standard errors are reported. Clustered standard errors, used in Study 2, are less appropriate here for two reasons. Firstly, with only five survey years, the number of clusters is well below the 30-50 typically required for cluster robust survey inference (Cameron and Miller, 2015). Secondly, clustering is most consequential when observations within clusters are highly correlated on the outcome; given five outcome classes, within-year correlation is more spread out than Study 2 which uses a binary response. Year fixed effects absorb baseline shifts in thermometer ratings across survey years.

Relax LASSO We refit a standard multinomial logistic regression (GLM, relaxed LASSO) on the LASSO-selected terms to obtain less biased coefficient estimates and standard errors. Relaxed LASSO coefficients are interpreted as log-odds relative to the neutral reference class (class 3, thermometer 40–59).

For the skip-gram models, the relaxed GLM uses a slightly stricter sparsity threshold of 1.5% (terms appearing in at least 1.5% of documents) rather than 1%, yielding 135 terms for the Democrat model and 123 for the Republican model. This adjustment is needed because the full 1% skip-gram DTM exceeds the parameter limit of the multinomial model. This is methodologically sound because the LASSO step, using the full 1% DTM, was the primary output for variable selection.

5.2 Study 2: Modeling Selection into Open-Ended Response

Outcome Variables: Open-Ended Response Indicators Our outcome variables (DVs) for Study 2 are four binary indicators, one per open-ended item, of whether a respondent provided usable text. ANES gates each open-ended question with a screener (e.g., "Is there anything in particular that you like about the Democratic Party?"), and only respondents who say "Yes" are presented with the text field. We construct each outcome by combining the screener and the text field: the indicator equals 1 if the respondent answered "Yes" *and* the corresponding text field contains non-empty content, and 0 if the respondent answered "No" *or* answered "Yes" but left the text field empty. Both 0-groups are pooled because both produce no analyzable text and therefore drop out of the Study 1 corpus in the same way. The resulting indicators `y_dem_like`, `y_dem_dislike`, `y_rep_like`, and `y_rep_dislike` capture selection into the Study 1 sample on each of the four items separately. Across years, response rates on the four items range between roughly 40% and 60%, motivating an explicit selection model.

Predictors: Demographic and Political Variables We use the demographic and political-attitude variables in the cleaned ANES data set as candidate predictors of selection. These include partisanship (`pid7`), ideological self-placement (`ideology`), perceived ideological position of each party (`dem_ideo`, `rep_ideo`), party-difference perception (`party_diff`), feeling thermometers (`dem_therm`, `rep_therm`), political-engagement items (`right_track`, `understand_issue`, `too_complicated`, `power_matters`, `dem_satis`), and demographics (`age`, `gender`, `race_eth`, `educ7`, `work_status`, `relg`, `marital_status`, `sexori`, `amer_parents`, `health_insur`, `reg_census`). To improve interpretability of the relaxed-LASSO output, the seven-point ordinal items (`pid7`, `ideology`, `dem_ideo`, `rep_ideo`, `educ7`) are collapsed into three levels with the middle category as the reference (e.g., pid7 → → Democrat / Independent / Republican; educ7 → → less than HS / some college / BA+). Listwise deletion is applied to the candidate predictor set prior to estimation. Note that the open-ended text itself is not used as a predictor here, because Study 2 models who reaches the open-ended sample in the first place.

Logistic LASSO We use a logistic LASSO to identify which demographic and political variables most strongly predict whether a respondent contributes usable text on each item. We use `cv.glmnet` with `family = "binomial"` and `alpha = 1`. LASSO is appropriate here because the candidate-predictor set is moderately large, contains a number of categorical variables that expand into many indicator columns under `model.matrix`, and includes attitudinal items that are plausibly correlated with one another (e.g., `pid7` with `ideology`, the two thermometers with `party_diff`). Penalized regression both selects a parsimonious subset and stabilizes estimation under this multicollinearity. We force in survey year (`year`) as a factor variable with zero penalty on its dummy coefficients via the `penalty.factor` argument. As in Study 1, this guarantees year fixed effects are present in every model, absorbing the year-to-year shifts in baseline non-response that follow from changes in survey administration mode and sample composition. Coefficients on the substantive predictors therefore describe within-year associations between respondent characteristics and selection.

We use `lambda.1se` as our regularization parameter. This yields a more parsimonious model than `lambda.min` while maintaining comparable predictive performance.

Relaxed LASSO We refit a standard logistic regression (GLM, relaxed LASSO) on the LASSO-selected variables to obtain less biased coefficient estimates and standard errors.

6 Results

6.1 Study 1: Predictive Language in Party Evaluations

LASSO variable selection

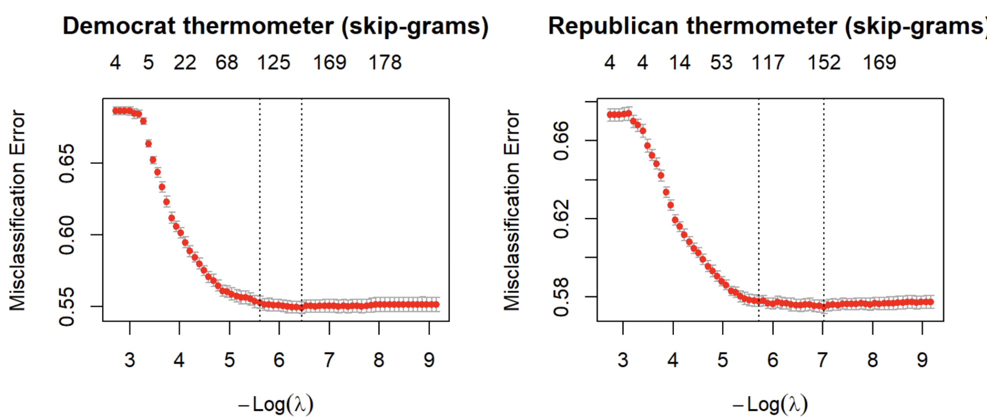

Figure 5: Thermometers for Both Parties

At `lambda.1se`, the Democrat skip-gram model selects 200 unique terms across all five thermometer classes, and the Republican model selects 188 terms. The number of selected terms varies substantially by class, with strong animosity (class 1) and favorability (class 5) having the most terms selected (136 and 126 for Democrats; 122 and 105 for Republicans), and neutral (class 3) and mildly unfavorable (class 2) classes having fewer (41 and 23 for Democrats; 68 and 40 for Republicans).

This is not surprising, as respondents with more extreme thermometer ratings would use more consistent and distinct language than those with moderate views, providing stronger text signals for LASSO to detect. The cross validation plots indicate that misclassification error plateaus well before `lambda-min`, meaning that `lamda.1se` captures the majority of predictive signal while remaining parsimonious.

Issue- and Value-Based Words Dominate

We now present word clouds of the top 25 LASSO-selected terms for animosity (class 1, thermometer 0–19) and favorability (class 5, thermometer 80–100) for both parties. Word size is proportional to the magnitude of the LASSO coefficient.

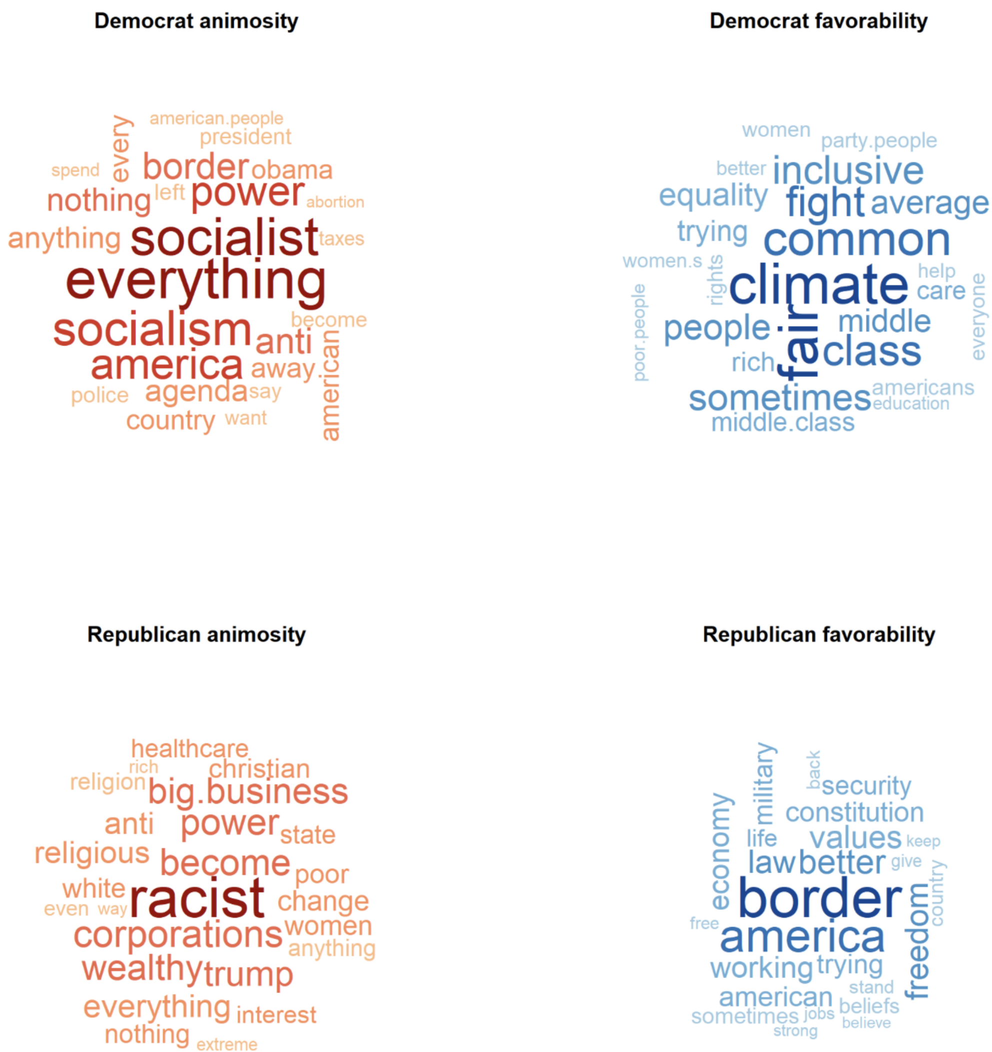

Figure 6: Word Clouds for Favorability and Animosity for Both Parties

Across both parties, issue-based and value-based language dominate the top predictors. For Democrat animosity, the strongest predictors are abstract value-based terms ("everything", "socialist", "socialism", "america") and policy-adjacent terms ("border", "illegal", "crime"). The presence of "socialist" and "socialism" as the dominant animosity predictors is consistent with an issue-based account of disliking — these terms reflect a policy orientation, not personal character traits. The term "everything" likely reflects a generalized frustration rather than a specific policy objection.

For Republican animosity, the pattern differs in character. The top predictor is "racist", followed by economic class-related terms ("corporations", "wealthy", "big.business", "poor", "rich") and identity-adjacent terms ("white", "religious", "christian"). Although "racist" is a trait-based term, the economic class language ("corporations", "big.business") clearly reflects policy concerns about economic inequality.

Notably, the dominant words are not the same across parties. Words associated with favorability are more specific and policy-related: Democrats are favored for climate, fairness and equality ("fair", "equal", "inclusive", "climate"), while Republicans are favored for national identity and rule-of-law values ("border", "america", "freedom", "law", "constitution"). Words associated with animosity are more abstract and value-laden. This asymmetry suggests that in-party favorability is grounded in concrete policy preferences, while out-party animosity reflects broader value conflicts — consistent with the issue-based account.

The chart below presents the top 10 LASSO coefficients for each of the four outcome classes, reaffirming the above results.

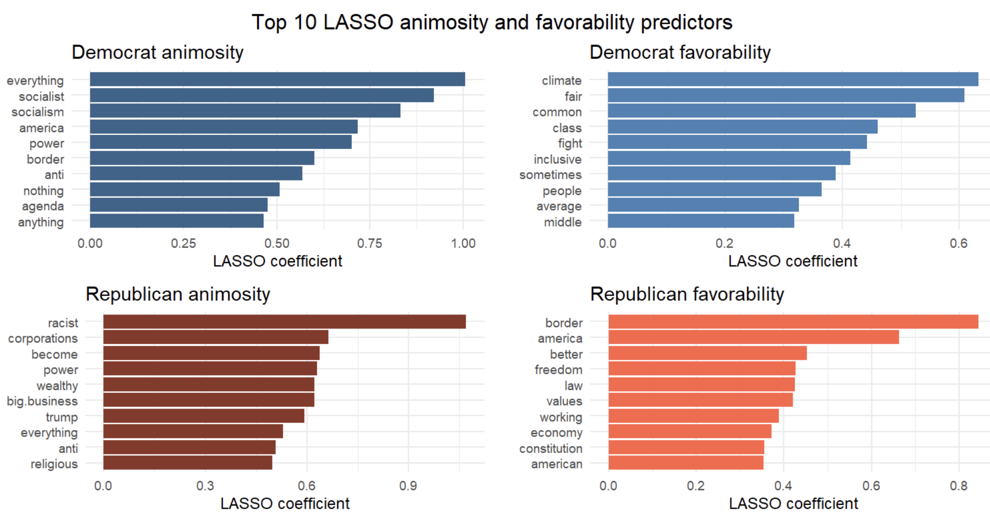

Figure 7: Top 10 LASSO Favorability and Animosity Predictors

Relaxed LASSO Confirms LASSO Findings

The relaxed LASSO (multinomial logistic regression on LASSO-selected terms) produces results consistent with the LASSO estimates. Coefficients are expressed as log-odds relative to the neutral class (thermometer 40–59). Figure 6 presents the top 10 predictors for each outcome class with 95% confidence intervals. Filled circles indicate terms significant at the 5% level.

The relaxed LASSO results reaffirm the LASSO findings. Terms with the largest positive log-odds for Democrat animosity (relative to neutral) include “socialist”, “everything”, “border”, and “illegal”. For Republican animosity, “racist”, “corporations”, “wealthy”, and “trump” show the largest effects. Favorability predictors are similarly consistent with the LASSO word clouds.

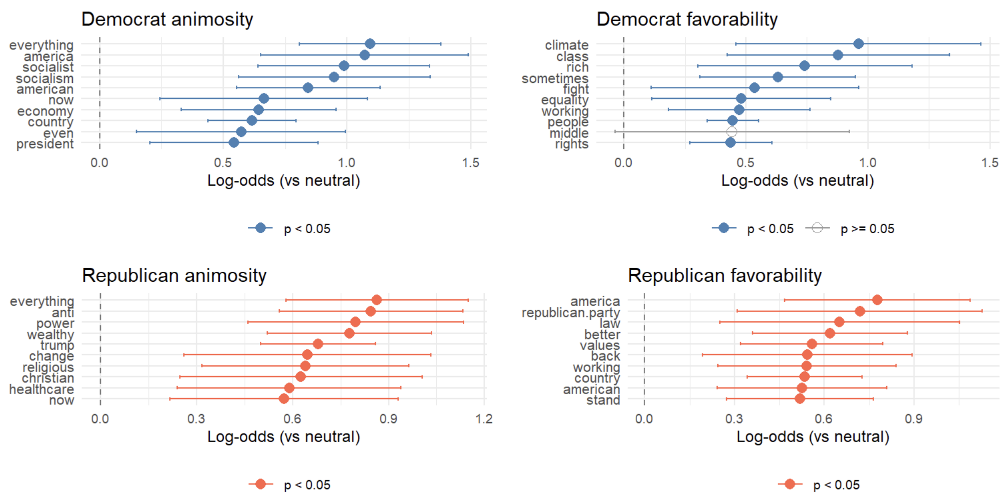

Figure 8: Relax LASSO Results

6.2 Study 2: Selection into Open-Ended Response

LASSO variable selection

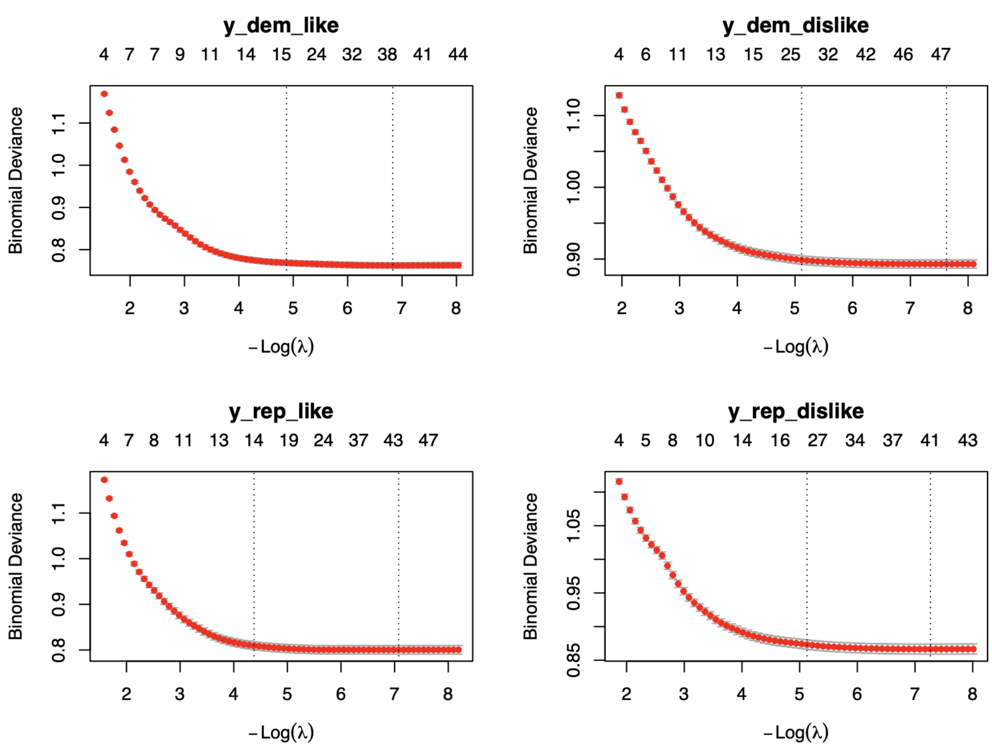

Figure 9: Cross-Validation Plots for the Four Per-Item Selection Models

For each of the four like/dislike items, we fit a binary logistic LASSO predicting whether the respondent provided usable open-ended text (coded 1 if `*_binary == 1` and the text field was non-empty; 0 if `*_binary == 5` or text was `NA`). Predictors include partisanship, ideology placements, party-difference perception, party thermometers, and the full demographic battery (age, gender, race/ethnicity, education, marital status, work status, religion, region, health insurance). Year is forced into every model via `penalty.factor = 0` so that selection is identified within election cycles, and the relaxed-LASSO step uses year-clustered robust standard errors (`sandwich::vcovCL`, HC1). The analytic sample after listwise deletion is N = 17,844, with adequate counts in every year (2008: 1,251; 2012: 4,456; 2016: 2,618; 2020: 5,772; 2024: 3,747).

At `lambda.1se`, the four models select different numbers of predictors: 15 for Democratic likes ($\lambda_{1se} = 0.0076$), 26 for Democratic dislikes ($\lambda_{1se} = 0.0060$), 14 for Republican likes ($\lambda_{1se} = 0.0126$), and 25 for Republican dislikes ($\lambda_{1se} = 0.0059$). The asymmetry between like and dislike models is striking — articulating dislikes draws in roughly 70% more predictors than articulating likes. The cross-validation plots show binomial deviance plateauing well before `lambda.min`, indicating that `lambda.1se` retains the bulk of predictive signal while keeping the models parsimonious.

Political Engagement, Partisanship, and Education Dominate

Three predictor families do most of the work across all four models. *Political engagement*, captured by `party_diff2` ("yes, there is a difference between the parties") and `party_more_consv2` (correctly identifying Republicans as the more conservative party), is among the strongest predictors throughout, with log-odds between 0.5 and 0.9 in every model. Respondents who perceive meaningful inter-party differences are markedly more likely to articulate why they like or dislike either party.

Partisan strength is the second consistent family. Both `pid7Dem` and `pid7Rep` (relative to independents) carry positive coefficients in nearly every model, indicating that identifying with either party, not the direction of identification — predicts response. Pure independents are the most likely to be silent. The exception is `y_rep_like`, where only `pid7Rep` is significant: essentially only Republicans articulate Republican likes.

Education is the third family and the single most consistent variable across all four models. `educ7BA_plus` is positive ($\beta \approx 0.5$) and significant at $p < 0.001$ in every per-item model, while `educ7NoHS` is consistently negative. College-educated respondents are roughly 60% more likely in odds-ratio terms to provide usable open-ended text than the high-school-or-some-college reference group, an effect of direct substantive concern for any analysis, including our Study 1, that conditions on the open-ended sample.

A clear asymmetry emerges between like and dislike models. Demographics, including age, race/ethnicity, gender, marital status, work status, drop out almost entirely from the like models but enter forcefully into both dislike models. Older respondents ($\beta \approx 0.013$–0.016 per year) are more likely to articulate dislikes; Asian (`race_eth3`) and Hispanic (`race_eth5`) respondents are consistently less likely than white non-Hispanics; women are slightly less likely than men. *Whether one articulates what one likes about a party is essentially a function of political knowledge and education; whether one articulates what one dislikes is also structured by demographic position.* (See full LASSO output in Appendix S2T1)

Relaxed LASSO with Year-Clustered SEs Confirms LASSO Findings

To make inference, we refit each LASSO-selected variable set as a multivariable logistic regression with standard errors clustered by year. The relaxed-LASSO estimates closely track the LASSO coefficients in sign and rank, and the dominant predictors- `party_diff2`, `educ7BA_plus`, `pid7Dem`, `pid7Rep`, and the in-party ideology placements (`dem_ideoLib` for Democrats; `rep_ideoCon` for Republicans) -remain significant at $p < 0.001$ in every model. As expected from coefficient debiasing, the relaxed-LASSO point estimates are slightly larger in absolute magnitude than the penalized LASSO estimates.

Discussion and Conclusion

Across two studies, we find that Americans mainly rationalize their likes and dislikes for the Democratic and Republican party in issue-based terms. We find little evidence of trait-based rationalization of partisan favorability or animosity. Though our analysis is only a descriptive, face-value analysis of our text mining results, our findings suggest American political polarization may be less affect-driven and more substantive ideology- or issue-driven than many scholars and commentators suppose. We also found that Americans describe their in-party in predominantly concrete, policy-specific terms (such as "climate" and "abortion"), while describing their out-party in more abstract, value-oriented terms (such as "socialist" and "racist"). This may suggest that partisans have less knowledge about out-partisans' substantive policy opinions than their in-party. In other words, they use partisanship labels as cues such that while they understand out-partisans have very different issue positions from themselves, they are not necessarily able to bring top-of-mind the specific policy disagreements without prompting. In contrast, it appears in-party favorability is driven substantive alignment with the party on specific policy issues.

Finally, we caution our findings by highlighting that our effective sample is biased toward people who are politically interested, highly educated, male, white, and partisan, as Study 2 shows. Our findings here also points to a long-standing criticism of using open-ended survey responses to assess public opinion. In particular, people who are less educated tend to give less information-rich and less articulate responses. However, this is not necessarily equal to them having no opinion, and can be simply an artifact of the open-ended survey format.

The biggest limitation of the current project is a lack of detailed conceptualization and operationalization of what counts as issue-based or trait-based partisan favorability. In future work, we would like to extend on our present analysis by constructing or utilizing a theory-driven codebook, such that the two types of favorability are more clearly delineated. In addition, we will manually code a small subset of training data from the raw open-ended responses and use large-language models to code the rest of our effective sample. Given that this method of coding would even-better account for contextual dependencies than skip-grams, we expect our findings to be better interpretable. Fundamentally, however, deeper exploration of the topic of causes of partisan favorability and animosity requires a deeper literature search of how trait-based and issue-based partisan stereotypes differ.

References

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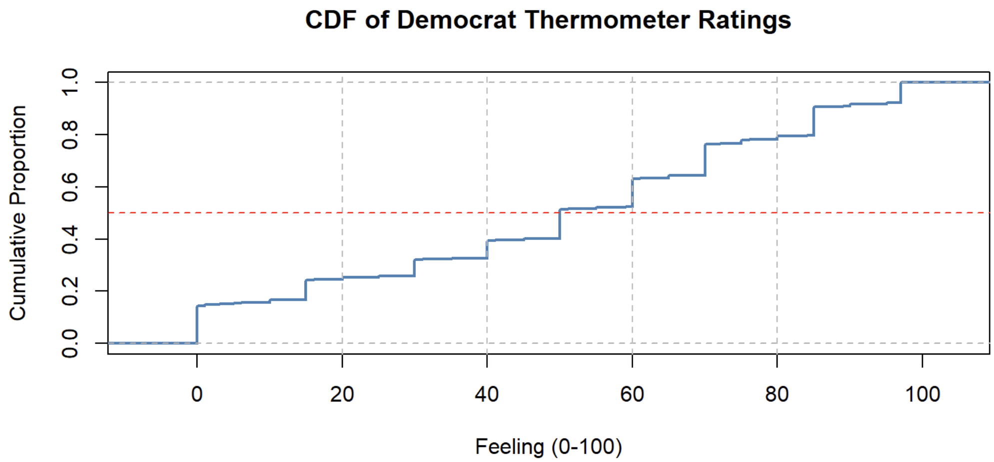
Appendices

6.3 Appendix S1G1



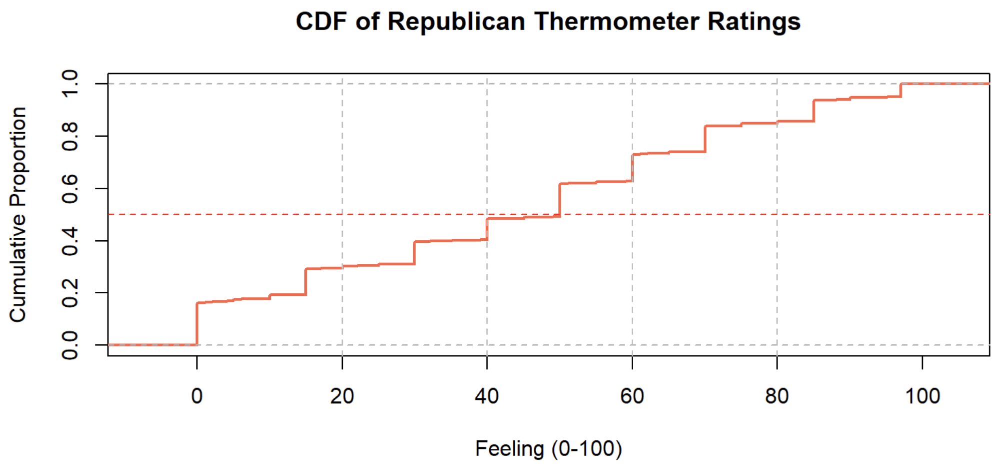


Figure 10: CDF of Thermometer Ratings

6.4 Appendix S1T1

Table 1: Distribution of Feeling Thermometer Outcome Classes (2008–2024, pooled).

	Democrat thermometer		Republican thermometer	
Class	N	%	N	%
1 — Highly unfavorable (0–19)	6375	24.6%	7649	29.5%
2 — Unfavorable (20–39)	2125	8.2%	2798	10.8%
3 — Neutral (40–59)	5083	19.6%	5811	22.4%
4 — Favorable (60–79)	6725	25.9%	5760	22.2%
5 — Highly favorable (80–100)	5644	21.7%	3891	15%

6.5 Appendix A1: Alternative Outcome: CSES Party Liking Scale

As an alternative to the ANES feeling thermometer, we re-estimated the skip-gram models using the CSES party liking scale (0–10), recoded into five parallel classes (0–1, 2–3, 4–6, 7–8, 9–10). The figure below compares the top 10 predictors for each class between the thermometer and CSES models.

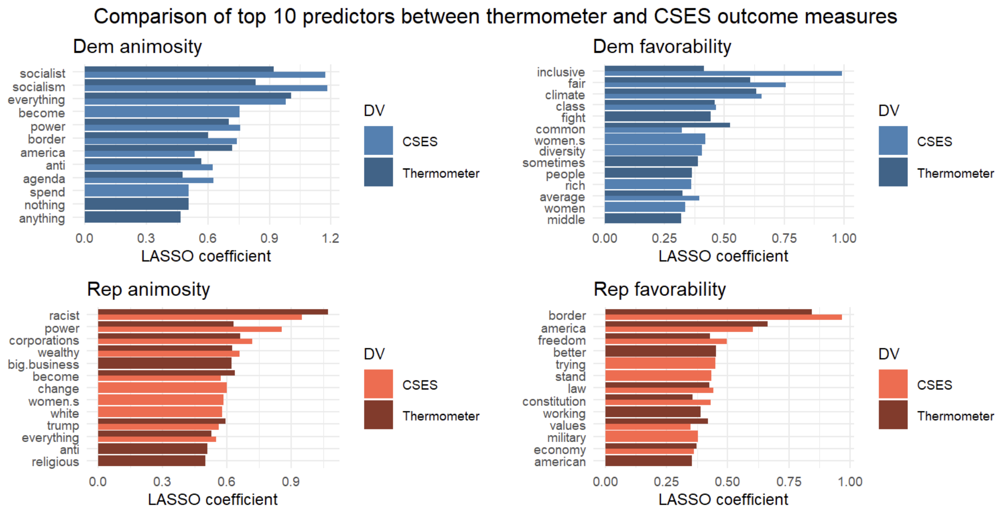


Figure 11:

Results are substantively consistent across both measures. One difference is that Republican favorability predictors under CSES include more specific policy terms ("military", "pro-life", "constitution") compared to the thermometer model, which is more dominated by national identity terms ("border", "america"). This suggests the CSES scale may capture an even more policy-grounded dimension of Republican liking.

6.6 Appendix A2: Partisan Stratification

To assess whether findings from the full sample are driven by in-party respondents, we re-estimated the skip-gram models on out-party samples only: Republicans rating Democrats (n = 6,526) and Democrats rating Republicans (n = 8,028). Due to a large class imbalance in out-party samples (approximately 60% of out-party raters fall in class 1, animosity), we use `type.measure = "deviance"` rather than `type.measure = "class"` for cross-validated lambda selection; the latter caused LASSO to select zero text terms by predicting all cases as the majority class.

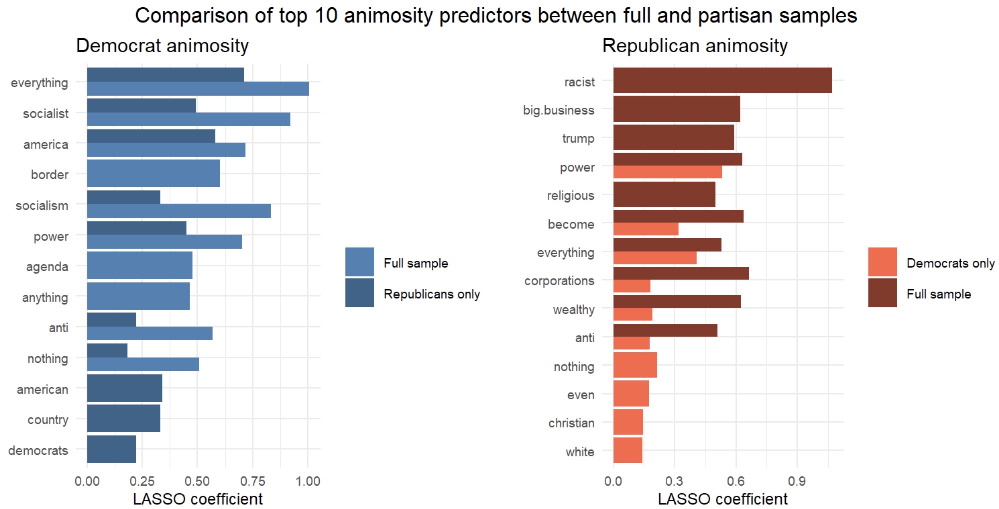


Figure 12:

We compare the top 10 animosity predictors (class 1) between the full sample and partisan-stratified models. Results are broadly consistent. The core animosity terms ("socialist", "everything", "border" for Democrats; "racist", "corporations", "wealthy" for Republicans) replicate across both samples.

6.7 Appendix A3: Alternative Text Representations: Stemming and N-Grams

We estimated parallel models using stemmed unigrams and bigram n-grams in addition to the primary skip-gram representation. We compare animosity (class 1) predictors between skip-grams and stems, and then between skip-grams and n-grams, for both Democrat and Republican thermometers.

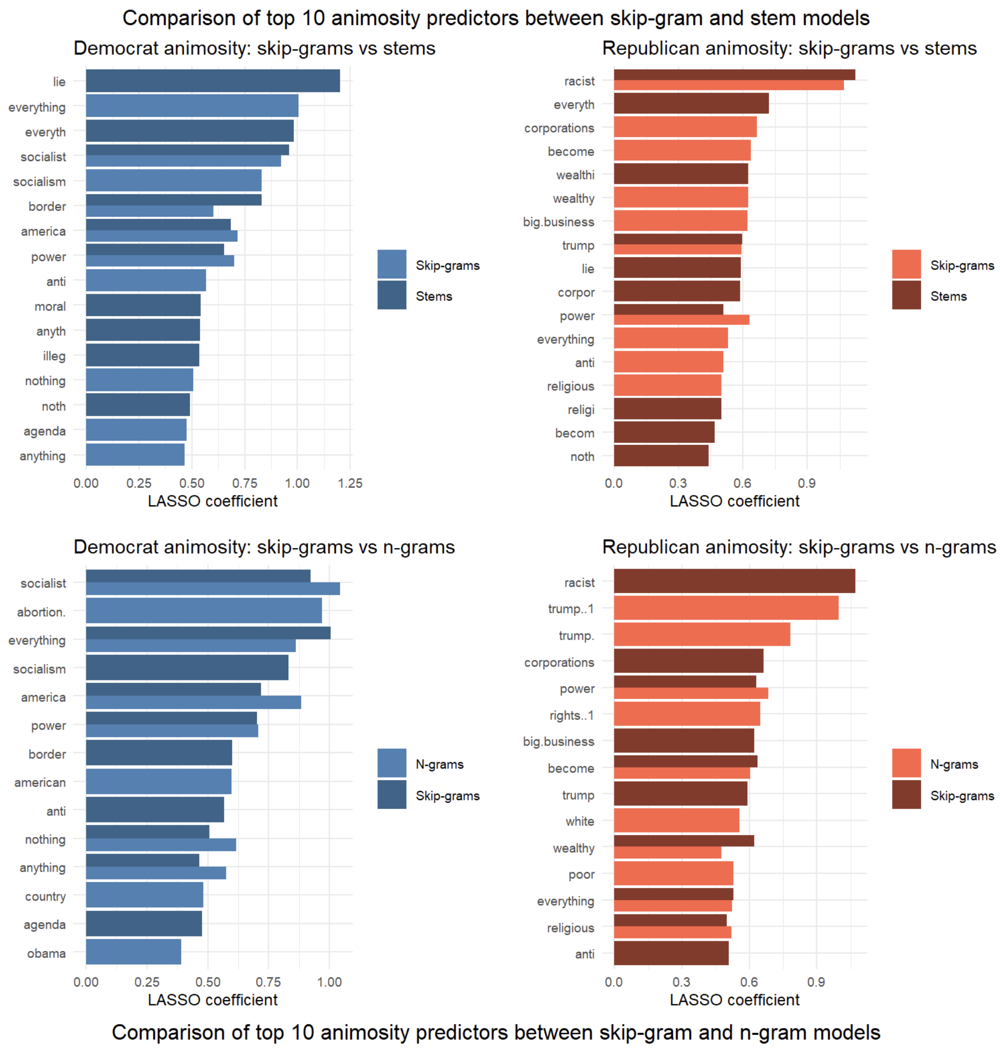


Figure 13:

Across both comparison figures, the substantive findings are consistent: the same core terms appear as top animosity predictors regardless of text representation. Skip-grams capture a few multi-word combinations ("big.business", "middle.class") not available to stems, and can produce more readable labels than stems. These comparisons support the robustness of the primary skip-gram findings.

6.8 Appendix A4: Fixed Effects Support EDA: Partisan Animosity Has Increased Over Time

The table below presents the year fixed effects on animosity (class 1) from all six models, with 2008 as the reference year. Conditional on controlling for text context, animosity toward Democrats increased sharply from 2008 through 2020 before stabilizing slightly in 2024. Animosity toward Republicans shows a steadily increasing trend from 2012 through 2024. The reduction in neutral ratings (class 3, negative coefficients across years in all models) is consistent with affective polarization literature documenting the decline of the political middle.

Table 2: Year fixed effects on animosity (class 1) relative to 2008.

	Year fixed effect (ref = 2008)			
Model	2012	2016	2020	2024
Democrat animosity	0.638	1.051	1.244	1.179
Republican animosity	0.177	0.098	0.258	0.327

6.9 Appendix S2T1

Table 3: Relaxed-LASSO logistic regressions predicting open-ended response. Coefficients are log-odds; year-clustered standard errors (HC1) in parentheses. "—" indicates the variable was not selected by LASSO at `lambda.1se` for that outcome. Reference categories: pid7 = Independent; ideology / dem_ideo / rep_ideo = Moderate; educ7 = HS or some college; gender = Male; race_eth = White non-Hispanic; relg = Protestant; reg_census = Northeast; year = 2008. Significance: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

	Dem like	**Dem dislike**	**Rep like**	**Rep dislike**	**Pooled**
(Intercept)	-2.52^{***}	-2.20^{***}	-3.46^{***}	-0.86^{*}	-2.18^{***}
	(0.111)	(0.244)	(0.098)	(0.349)	(0.214)
Year fixed effects (forced in)					
year = 2012	-0.555^{***}	0.515^{***}	0.242^{***}	-0.524^{***}	-0.401^{***}
	(0.022)	(0.015)	(0.014)	(0.042)	(0.050)
year = 2016	-20.1^{***}	-19.0^{***}	-19.0^{***}	-20.3^{***}	-21.4^{***}
	(1.12)	(1.12)	(1.12)	(1.12)	(1.13)
year = 2020	-0.552^{***}	0.611^{***}	0.226^{***}	-0.691^{***}	-0.297^{***}
	(0.039)	(0.037)	(0.034)	(0.058)	(0.074)
year = 2024	-0.298^{***}	0.699^{***}	0.387^{***}	-0.416^{***}	-0.130^{*}
	(0.037)	(0.024)	(0.032)	(0.067)	(0.065)
Party thermometers					
Dem. thermometer	0.033^{***}	-0.026^{***}	-0.014^{***}	0.004^{***}	—
	(0.001)	(0.002)	(0.001)	(0.001)	
Rep. thermometer	-0.019^{***}	—	0.026^{***}	-0.033^{***}	-0.011^{***}
	(0.001)		(0.001)	(0.002)	(0.001)
Party identification (ref: Independent)					
Democrat	0.696^{***}	0.309^{**}	0.122	0.353^{*}	1.10^{***}
	(0.093)	(0.106)	(0.081)	(0.159)	(0.090)
Republican	0.319^{***}	0.371^{***}	0.578^{***}	0.274^{***}	1.11^{***}
	(0.077)	(0.085)	(0.036)	(0.060)	(0.090)
Political engagement					
Party diff. (yes)	0.589^{***}	0.570^{***}	0.639^{***}	0.680^{***}	0.915^{***}
	(0.081)	(0.051)	(0.094)	(0.116)	(0.047)
Party diff. (DK)	0.063	0.435	0.764^{***}	0.334	0.330^{**}
	(0.563)	(0.624)	(0.117)	(0.419)	(0.101)
Reps. more cons.	0.137	0.637^{***}	0.457^{***}	0.616^{***}	0.494^{***}
	(0.097)	(0.105)	(0.104)	(0.061)	(0.072)
Both same / DK	-0.245	0.171	0.090	0.036	-0.048

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Table 3 continued from previous page

	Dem like	**Dem dislike**	**Rep like**	**Rep dislike**	**Pooled**
	(0.140)	(0.098)	(0.130)	(0.068)	(0.162)
Right track (wrong)	—	0.191^{**}	—	—	-0.156^{**}
		(0.073)			(0.060)
Ideology placements (ref: Moderate)					
Self: liberal	0.286^{***}	0.204^{***}	-0.177^{**}	0.233^{***}	0.417^{***}
	(0.065)	(0.056)	(0.057)	(0.066)	(0.083)
Self: conservative	0.024	0.476^{***}	0.415^{***}	0.278^{***}	0.740^{***}
	(0.081)	(0.082)	(0.062)	(0.018)	(0.053)
Place Dem. as liberal	0.389^{***}	0.313^{***}	0.702^{***}	0.573^{***}	0.524^{***}
	(0.046)	(0.058)	(0.031)	(0.024)	(0.056)
Place Dem. as cons.	0.315^{**}	0.181^{***}	0.165	0.375^{*}	0.472^{***}
	(0.110)	(0.045)	(0.140)	(0.149)	(0.062)
Place Rep. as liberal	0.365^{**}	0.228^{***}	0.218	0.184^{***}	0.329^{*}
	(0.121)	(0.056)	(0.132)	(0.017)	(0.141)
Place Rep. as cons.	0.927^{***}	0.837^{***}	0.762^{***}	0.510^{***}	0.998^{***}
	(0.099)	(0.110)	(0.076)	(0.029)	(0.139)
Demographics					
Age	—	0.013^{***}	—	0.016^{***}	0.025^{***}
		(0.001)		(0.002)	(0.003)
Gender: Female	—	-0.287^{***}	-0.192^{***}	-0.299^{***}	-0.340^{***}
		(0.071)	(0.024)	(0.060)	(0.052)
Gender: Other	—	0.911^{***}	1.62^{***}	-0.289^{***}	0.498^{***}
		(0.046)	(0.108)	(0.028)	(0.104)
Race: Black	—	-0.138^{***}	—	-0.161^{*}	0.194
		(0.038)		(0.074)	(0.108)
Race: Asian/PI	—	-0.438^{***}	—	-0.547^{***}	-0.438^{**}
		(0.040)		(0.133)	(0.147)
Race: Native	—	−0.095	—	0.280	0.466^{**}
		(0.281)		(0.259)	(0.176)
Race: Hispanic	—	-0.203^{***}	—	-0.207^{**}	−0.006
		(0.015)		(0.079)	(0.106)
Race: Other/multi	—	0.183^{***}	—	0.086	0.188^{*}
		(0.036)		(0.084)	(0.093)
Educ: <HS	−0.151	-0.435^{***}	-0.220^{***}	-0.482^{***}	−0.182
	(0.100)	(0.061)	(0.039)	(0.047)	(0.167)
Educ: BA+	0.505^{***}	0.628^{***}	0.551^{***}	0.603^{***}	0.558^{***}
	(0.100)	(0.028)	(0.067)	(0.043)	(0.077)
Marital: Never married	—	-0.053^{***}	—	−0.080	—
		(0.013)		(0.055)	
Marital: Divorced	—	-0.211^{***}	—	-0.197^{***}	—
		(0.049)		(0.002)	
Marital: Separated	—	0.172^{*}	—	−0.049	—
		(0.081)		(0.225)	
Marital: Widowed	—	0.009	—	-0.214^{***}	—
		(0.047)		(0.049)	
Marital: Partnered	—	−0.050	—	-0.166^{***}	—
		(0.049)		(0.047)	
Work: Not employed	—	−0.051	—	—	—
		(0.094)			
Work: Retired	—	-0.131^{*}	—	—	—

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Table 3 continued from previous page

	Dem like	**Dem dislike**	**Rep like**	**Rep dislike**	**Pooled**
		(0.053)			
Work: Homemaker	—	−0.080	—	—	—
		(0.118)			
Work: Student	—	0.424***	—	—	—
		(0.114)			
Religion: Catholic	—	—	—	−0.018	—
				(0.060)	
Religion: Jewish	—	—	—	−0.013	—
				(0.090)	
Religion: Other/none	—	—	—	0.135***	—
				(0.030)	
Region: N. Central	—	−0.158*	—	0.020	—
		(0.080)		(0.029)	
Region: South	—	−0.032	—	−0.036	—
		(0.045)		(0.032)	
Region: West	—	0.145***	—	0.188***	—
		(0.019)		(0.057)	
Health insurance: No	−0.199*	−0.134	—	−0.089	−0.221
	(0.081)	(0.108)		(0.159)	(0.134)
Observations	17,844	17,844	17,844	17,844	17,844
LASSO selected (`lambda.1se`)	15	26	14	25	24
λ_{1se}	0.00763	0.00601	0.01256	0.00593	—